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Nose Pressure Distribution (rev.4)

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Introduction

This memo addresses the design of a rocket nose shape having a prescribed pressure distribution in incompressible flow. The idea is to delay vortex separation to a point as far aft as possible to mitigate induced roll moments and roll lock-in.

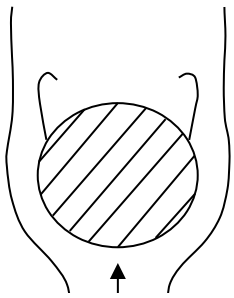
Motivation

Consider a sounding rocket subject to both earth-fixed (winds and gusts) and body-fixed (fin and thrust misalignments) perturbations. In general, the rocket is rolling. Viewed dynamically in a non-rolling frame, the response to the earth-fixed perturbations appears to not be influenced at all by the roll rate because its roll moment of inertia is of order of 1% of its pitch/yaw moments of inertia. Even though the rocket dynamics are not significantly influenced by roll, the response to body-fixed misalignments appears in a non-rolling frame to be driven by sinusoidal torques whose frequency is the roll rate.

Trouble can come when the roll rate equals the pitch/yaw natural frequency. In the steady state case, the response amplitude becomes very large because sounding rockets have very low pitch/yaw damping. In electrical engineering parlance, they would be said to have a high Q. Usually this resonance condition is a transient, not a steady state event so that the very large response amplitudes are not observed.

Nonetheless, higher angles of attack are commonly seen near resonance. If the angle of attack α exceeds a modest threshold, the cross-plane flow field near the tail looks like:

See the pair of vortices, one on each side, created from vorticity shed from the boundary layer. As we move this cross-flow picture further aft, the vortices grow stronger fed by more vorticity from the separated boundary layer.



U sinα

When the separated vortex pair reaches the tail fins, bad things start to happen. Vorticity induces an angle of attack distribution over the fins that integrates to a roll torque. This torque can exceed the roll driving torque from fin cant angles, and thus lead to roll lock-in.

Pressure Distribution

We can now state the reason why the external pressure distribution over a rocket nose matters. If the external pressure profile is smooth and continuously decreasing, the boundary layer can be inhibited from separation. Adverse pressure gradients in which the pressure increases moving aft facilitate boundary layer separation and the development of a separated vortex pair as sketched above. There are two components of pressure gradient: that in the circumferential direction and that in the longitudinal direction, and three velocity components, those in the axial, radial and circumferential directions. The circumferential velocity q_θ is, in the absence of separated vorticity, given by[3]

$$q_\theta = 2 U_\infty \sin\alpha \sin\theta.$$

The corresponding circumferential pressure distribution for pure cross plane flow is given by

$$Cp = \sin^2\alpha (1 - 4 \sin^2\theta),$$

where Cp = Pressure coefficient $\equiv (p - p_\infty) / \frac{1}{2} \rho U_\infty^2$,
p = Static pressure,
ρ = Mass density
U = Flight speed,
$( )_\infty$ = Conditions at a great distance from the body,
α = Angle of attack, and
θ = Circumferential angle.

It can be seen that the circumferential pressure distribution is a maximum on the leeward and windward meridians. Also, this velocity component and pressure distribution does not depend in any sense on body shape or size. There's nothing for a designer to do but address flight speed and angle of attack. But, as we shall see in the next section, the meridional shape has a significant bearing on the longitudinal pressure distribution.

Munk[4] Airship Theory

The basic idea here is to represent the incompressible flow external to body of revolution at zero α by a superposition of singular elementary sources along the body centerline. Note that negative sources, or sinks, are included. A source is defined by its strength Q and location. Consider a source continuously emitting Q cubic feet per second of an incompressible fluid. Conservation of mass requires that the radial velocity q_R through a control surface at distance r from the source be given from

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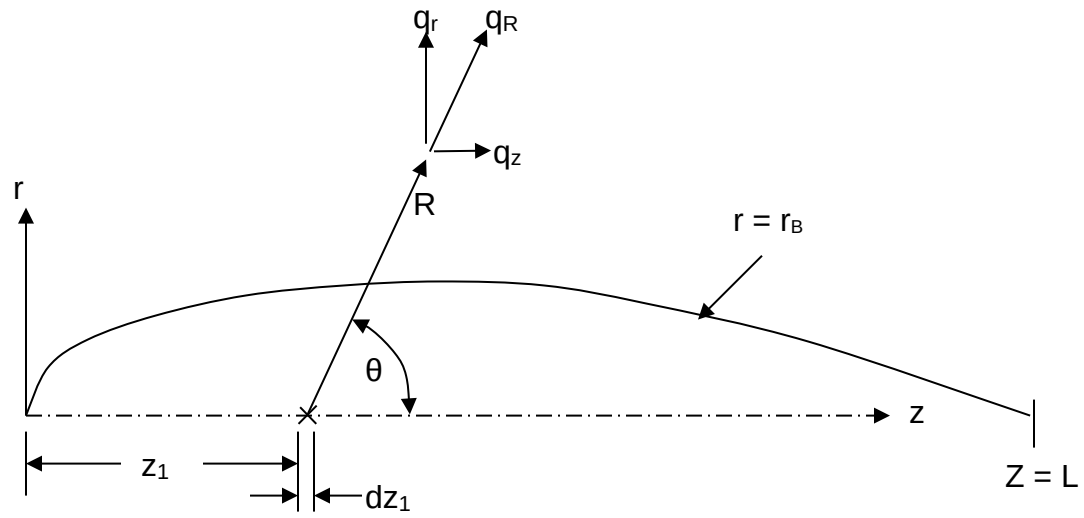
[3] M. Rauscher, "Introduction to Aeronautical Dynamics", John Wiley & Sons, Inc., New York, 1953, p. 241

[4] M. M. Munk, "Note on the Pressure Distribution over the Hull of Elongated Airships with CircularCross Section", N.A.C.A. TN-192, March 1924.

$$Q = 4 \pi r^2 q_R, \text{ or } q_R = Q / 4 \pi r^2$$

Such sources are called singular because of their behavior as r becomes vanishingly small.

Next, consider a body of revolution as sketched below:



Here the body mold line (assumed to be continuous) is given by $r = r_B(z)$. The local source strength at $z = z_1$ is
$\sigma(z_1)\, dz_1$. Then,

$$q_R = \sigma(z_1)\, dz_1 / 4 \pi R^2 = \sigma(z_1)\, dz_1 / 4 \pi ((z - z_1)^2 + r^2).$$

Also,

$$\sin \theta = r / ((z - z_1)^2 + r^2)^{1/2} \quad \text{and} \quad \cos \theta = (z - z_1) / ((z - z_1)^2 + r^2)^{1/2}$$

Putting these pieces together results in

$$q_r(z, r) = (1 / 4 \pi) \int_0^L r\, \sigma(z_1)\, dz_1 / ((z - z_1)^2 + r^2)^{3/2}, \text{ and}$$

$$q_z(z, r) = (1 / 4 \pi) \int_0^L (z - z_1)\, \sigma(z_1)\, dz_1 / ((z - z_1)^2 + r^2)^{3/2}.$$

The source intensity distribution is determined by the tangency boundary condition at the body surface. The flow from the z axis source distribution can be thought of as nearly radial because the q_z contributions from nearby regions tend to

cancel each other out, an assumption valid except near the nose stagnation point. Then the radial velocity is

$$2 \pi r \, dz_1 \, q_r(z, r) = \sigma(z_1) \, dz_1, \text{ or}$$

$$q_r(z_1, r) = \sigma(z_1) / 2 \pi r$$

Also, the longitudinal velocity is approximately U_∞, the free stream velocity. Then, if there is no flow through the surface,

$$q_r(z_1, r) / U_\infty = dr_B/dz, \text{ or}$$

$$\sigma(z_1) = 2 \pi U_\infty r_B \, dr_B/dz.$$

Pressure Coefficient

From Bernoulli's law (obtained by integrating the momentum equation along a streamline)

$$p + \tfrac{1}{2} \rho U^2 = p_\infty + \tfrac{1}{2} \rho U_\infty^2$$

This can be rewritten as

$$Cp \equiv (p - p_\infty) / \tfrac{1}{2} \rho U_\infty^2 = 1 - (U / U_\infty)^2.$$

Note that Cp = 1 at a stagnation point. Expanding this gives

$$Cp = 1 - (U_\infty^2 + 2 q_z(z, r) U_\infty + q_z(z, r)^2 + q_r(z, r)^2 + q_\theta^2) / U_\infty^2$$

For a slender body,

$$Cp = - 2 q_z(z, r) / U_\infty - 4 \sin^2\alpha \, \sin^2\theta$$

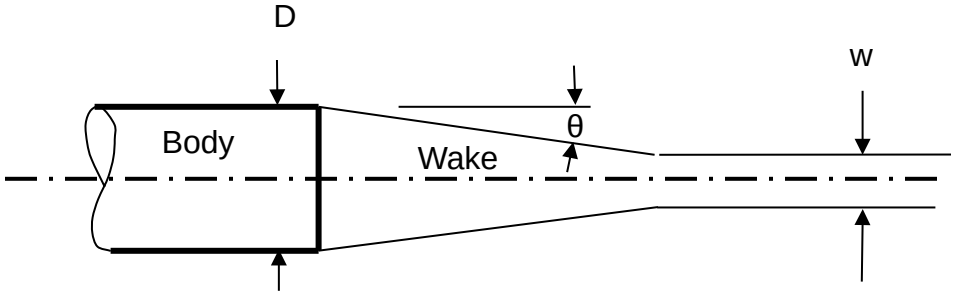
The last term above is just that due to cross plane (circumferential) flow, and, as already noted, because it offers no opportunity to the designer, will be ignored hereafter. Then,

$$Cp (z, r_B) = - (1 / 2 \pi U_\infty) \int_0^L (z - z_1) \, \sigma(z_1) \, dz_1 / ((z - z_1)^2 + r_B^2)^{3/2}, \text{ or}$$

$$Cp (z, r_B) = - \int_L (z - z_1) \, r_B(z_1) \, dr_B/dz \, (z_1) \, dz_1 / ((z - z_1)^2 + r_B^2)^{3/2}$$

Body Profile

It might be thought that the body profile would not be an issue, but it is. The thing is, if one terminates the body profile at the base of the body, significant error will be introduced to regions well forward of the base. Consider a streamline surface covering the body just outside the boundary layer. Absent any thrust, at the body base this flow separates and converges along a conical shape until the wake core is reached. The wake core can be modeled as a vey long cylinder. With thrust, a more complex wake shape emerges. However, we do not need a highly accurate description of the near body wake so the non-thrusting wake profile will be used for the general analysis. When the wake after body is included in the body profile, Cp estimated over the forebody has greatly improved accuracy.



The first relationship we obtain is that for Impulse – Momentum in the cylindrical after wake. Consider the element of mass added to the cylindrical wake during time dt. Its mass is $\rho\frac{\pi}{4}w^2Udt$. Its change in velocity relative to the body is $U - U_W$. The resulting change in momentum must be equal to the drag impulse applied by the body to the air. Then,

$$Drag * dt = \rho\frac{\pi}{4}w^2Udt * (U - U_W), \text{ or}$$

$$\frac{1}{2}C_D\left[\frac{D}{w}\right]^2 = \frac{U - U_W}{U} \text{ where}$$

C_D = Forebody drag coefficient (including base drag) based on maximum body cross section area,
U = Free stream velocity,
U_W = Cylindrical wake velocity relative to the body,
D & w =Body and cylindrical wake diameters, and
θ = Wake profile half angle.

The second relationship is basically that due to Bernoulli, neglecting any entrainment of air from the body base until the cylindrical after wake is reached. Then assuming that the wake pressure has recovered to its free stream value by the time the wake has reached its cylindrical state,

$$p_B = p + \frac{1}{2}\rho U_W^2, \text{ or}$$

$$\left[\frac{U_W}{U}\right]^2 = C_{pBASE} \text{ where}$$

p_B =Base pressure,
p = Free stream pressure,
C_{pBASE} = Base pressure coefficient, and
ρ = Air mass density.

These two equations may be solved for the cylindrical wake diameter:

$$w = D\sqrt{\frac{C_D}{2(1-\sqrt{C_{pBASE}})}}$$

The remaining parameter to be established is the conical fore wake convergence angle. For the moment, lacking better insight, we take $\theta = 0.1$ radian.

Karman-Tsien Compressibility Correction

The Munk theory described above applies to a slender body in incompressible flow. The next step is to adjust this incompressible pressure for the effects of compressibility. We note that such corrections only apply to subcritical Mach numbers. For any body, the critical Mach number is the lowest free stream Mach number for which the Mach number anywhere on the body surface is sonic. The Karman-Tsien compressibility correction is a semiempirical formula used to estimate the compressible Cp given an estimate of the incompressible Cp_0. The Karman-Tsien formula is[3]:

$$C_P = \frac{C_{P0}}{\sqrt{1-M_\infty^2} + \frac{C_{P0}}{2}(1-\sqrt{1-M_\infty^2})}$$

This correction factor can be used to find the pressure distribution in compressible flow at different free stream Mach numbers, M_∞. According to Felkel[3]

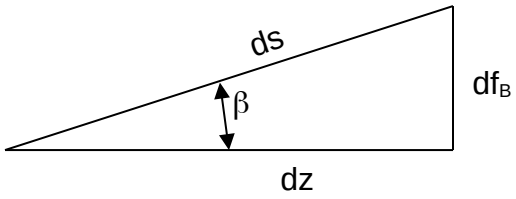
$$\frac{p^*}{p_\infty} = \left[\frac{(1/2)(\gamma+1)}{1+(1/2)(\gamma-1)M_\infty^2}\right]^{\frac{-\gamma}{\gamma-1}}$$

Here, γ = Ratio of specific heats,
M_∞ = Free stream Mach number, and
p* = Critical surface pressure.

Axial Force

In the process of estimating detailed loads, or when developing a drag budget for an entire vehicle, it becomes necessary to find the axial force resulting from the nose pressure distribution. First, consider an infinitesimal slice of nose of radius r_B. Now, if

$\tan \beta = dr_B/dz$



the total axial force acting on element of body surface is

$$d \text{ Total Axial Force} = r_B \, d\theta \, ds \, p \sin \beta \, .$$

Integrating around the full 2π of the slice, and noting that

$$dz = ds \cos \beta,$$

so that $d \text{ Total Axial Force} = 2 \pi r_B \, p \tan \beta \, dz$

Since $\tan \beta = dr_B/dz,$

we have that $d \text{ Total Axial Force} = 2 \pi p \, r_B \, dr_B = \pi p \, dr_B^2$

Next, recall that $p = p_\infty + q \, Cp,$

where $q = \text{Dynamic Pressure} = \frac{1}{2} \rho U_\infty^2.$

Then, $d \text{ Total Axial Force} = \pi (p_\infty + q \, Cp) \, dr_B^2.$

There are two terms in the above equation. The first is Hydrostatic and the second is Hydrodynamic.

- Hydrostatic Force $= p_\infty$ Base Area, and
- Hydrodynamic Force $= 2 \pi q \int_0^L Cp \, r_B \, (dr_B/dz) dz.$

The hydrostatic force is commonly ignored when describing drag because it vanishes when integrated over the entire body. However, when estimating interior loads, it can be important. The latter eq. must be numerically integrated, including the Karman – Tsien compressibility corrections, to find the total axial force acting on the forebody.

Numerical Approach

Now, suppose we have the body profile described in terms of radii r_B at a number of evenly spaced axial locations z. The next step is to estimate the derivative, dr_B/dz. Suppose we have three body radii, $r1$, $r2$ and $r3$ at three body stations $z1$, $z2$, and $z3$. Fit a parabolic curve through these points and differentiate it. Then, if the middle point

corresponds to z_2,

$$r_B(z_2)\, dr_B/dz\,(z_2) \;=\; r2\,(r3 - r1) \,/\, (z3 - z1)$$

This model has been coded into an Excel spread sheet called MUNKSHIP2.xls. Trapezoidal integration was used with the nose broken into many pieces for integration. This gives reasonable results for body regions remote from a blunt afterbody. The code does not model any wake or rocket motor exhaust plume.

This is straightforward except for the first (nose tip) body element. For a pointed nose tip, Cp takes on its full isentropic stagnation value at the very point. Assume that upon leaving the point the flow immediately expands to its conical value. Then letting the tip be subscripted "1" and the base of the first body element be subscripted "2". Then the Hydrodynamical Force on the first body element is:

- Nose Tip Hydrodynamic Force $= 2\,\pi\, q\, Cp_2\, (dr_B/dz)_2 \int_0^{z_2} r_B\, dz.$

After integration, we have that:

- Nose Tip Hydrodynamic Force $= \pi\, q\, Cp_2\, r_{B2}^2.$

The interesting thing about subsonic Nose Tip Hydrodynamic Axial Force is that it can be negative, i.e., the net axial force can be negative as in thrust.

A typical conical nose 36" long with a 6" base diameter followed by a cylindrical afterbody was chosen as a sample problem. Figure 1 below shows its longitudinal pressure distribution. Note especially the relatively large adverse pressure gradient beginning at the base of the nose cone caused by airflow speeding up to get around the cone-cylinder corner.

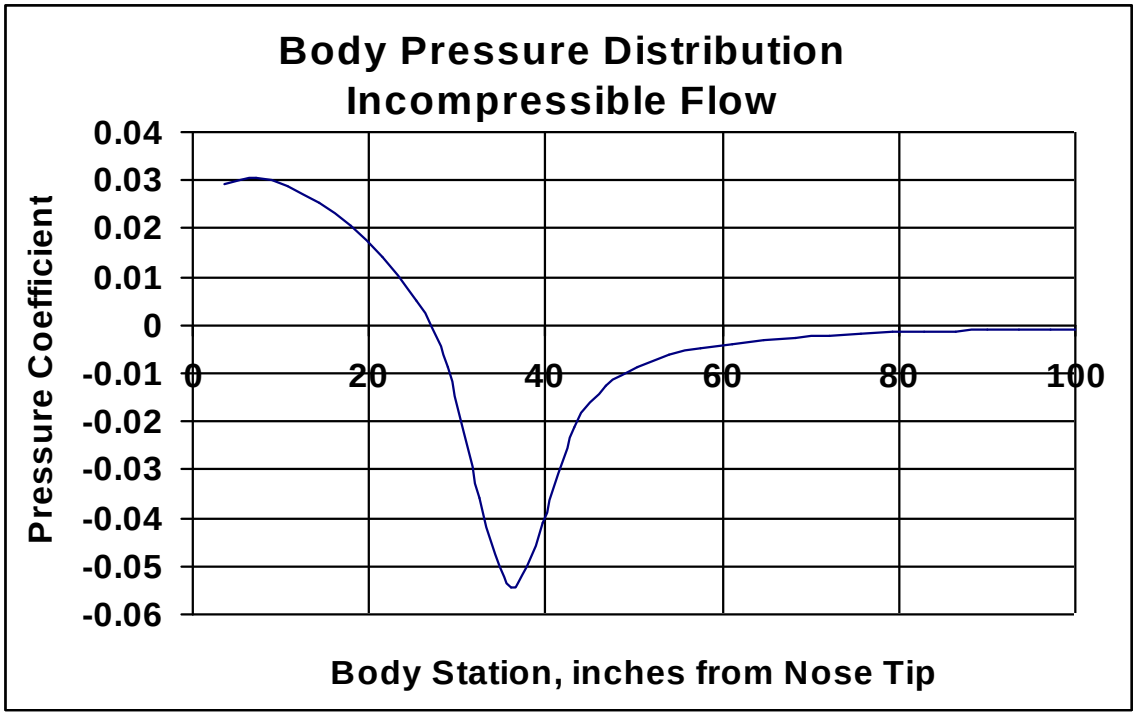


Figure 1

It was quickly discovered that for most nose shapes there was a large negative pressure coefficient bottoming out at nearly -0.06 around the base of the nose leading to a large adverse pressure gradient over the cylindrical afterbody. From the standpoint of lateral vortex separation this is not good. Manipulation of the individual body radii showed that while this region of negative pressure coefficient around the nose base could not be eliminated, it could be significantly mitigated. Figure 2 below shows the pressure distribution around one of the more promising nose profiles. The adverse pressure gradient over the cylindrical afterbody is still present, but is only half what it would have been without mitigation.

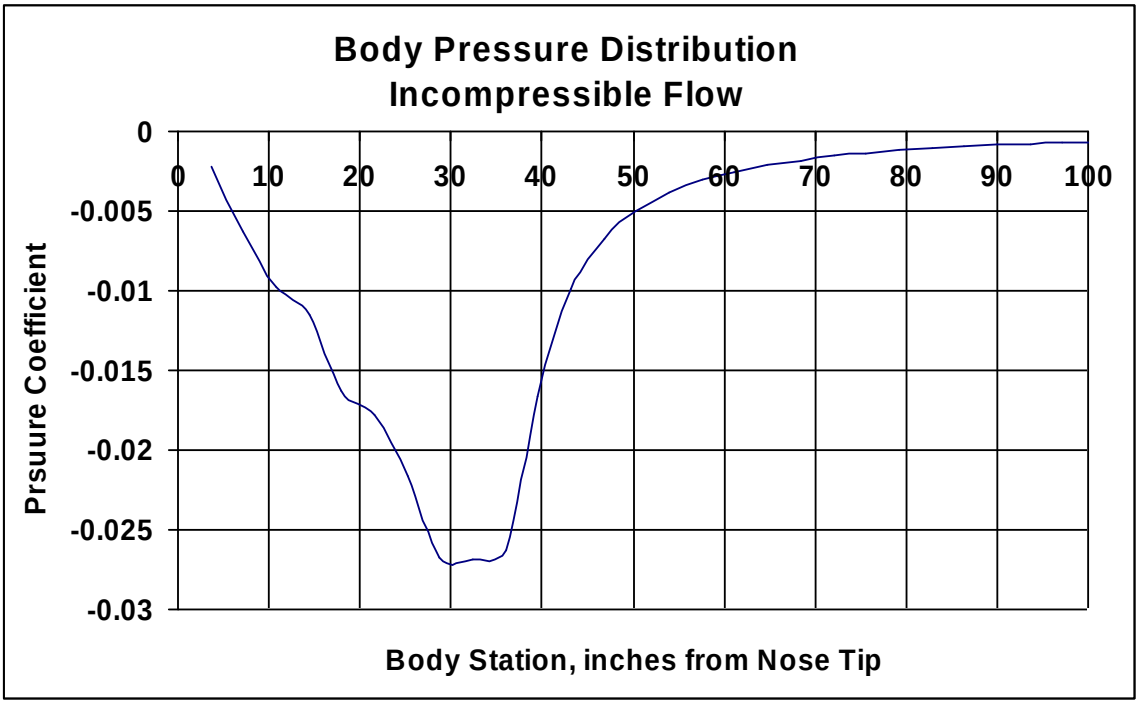


Figure 2

The body profile giving this pressure distribution, labeled rB, is shown in Figure 3 below. For purposes of comparison, a power law profile, labeled zn, is also shown for

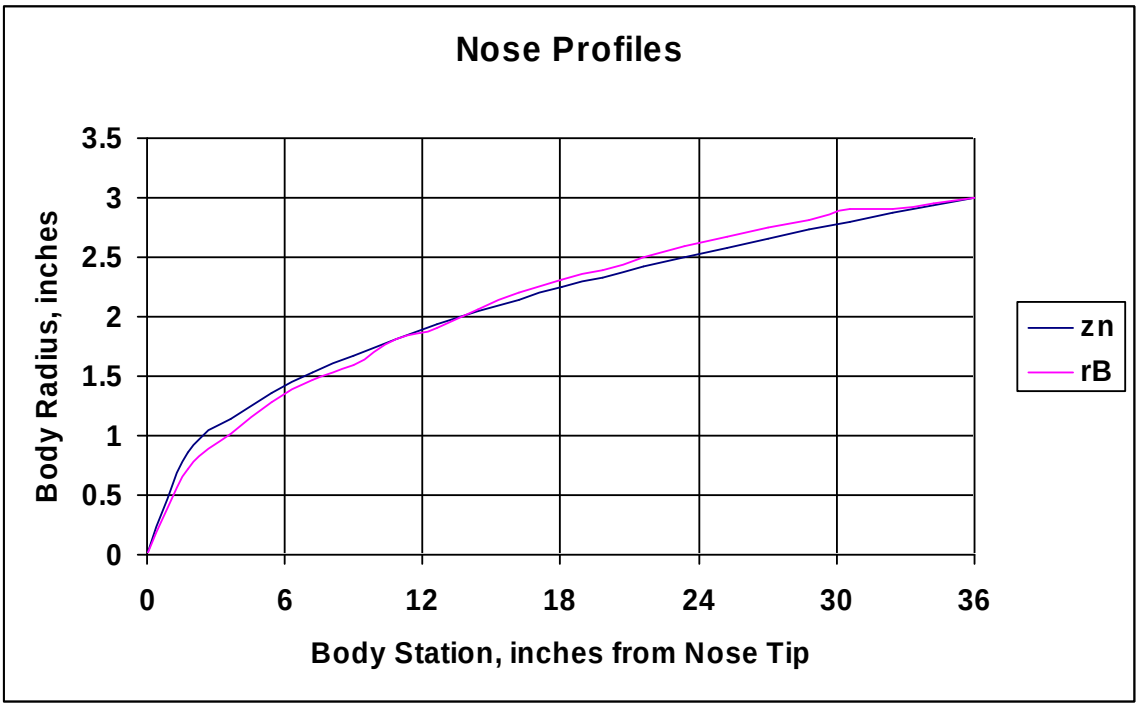


Figure 3

n = 0.42. While the rB curve is wriggly due to numerical noise it seems that such a power law profile might be a satisfactory practical solution to the problem of mitigating lateral vortex separation. For reference purposes, a table of radial ordinates is shown below:

Body Station, inches	$Z^{0.42}$, inches	rB, inches
0	0	0
3.6	1.064	1.010
7.2	1.454	1.466
10.8	1.745	1.816
14.4	1.986	2.065
18.0	2.196	2.307
21.6	2.384	2.506
25.2	2.555	2.674
28.8	2.713	2.815
32.4	2.861	2.911
36.0	3.000	3.000

Note that the column labeled $Z^{0.42}$ has been scaled to provide body radii in inches.